

Technical Document #1045: Retrieval Augmented Generation - Comprehensive Overview

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Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Vector databases store dense embeddings for semantic search.
- Re-ranking models score retrieved documents for relevance to the query.
- Hybrid search combines keyword-based and semantic search.